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Paleomagnetism of Pliocene Basalts from the Southwestern U.S.A.*

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Abstract

Late Pliocene basaltic lavas in New Mexico and Arizona are investigated paleomagnetically and petrographically. Detailed studies did not indicate any systematic difference in the magnetic or petrographic properties between normal and reversed rocks. Other evidence supports the idea of field reversals. The virtual geomagnetic poles determined from these lavas are grouped about the present geographic pole, which is in agreement with the paleomagnetic results for Pliocene rocks from other regions.

1. Introduction

Geophysicists and geologists now widely accept the idea that the geomagnetic field has reversed itself in the geological past. This hypothesis has become more and more convincing as additional paleomagnetic and rock magnetic data have accumulated. However, there still remains the possibility of naturally occurring self-reversal of the magnetic polarity in some rocks. Very few cases have been reported in which the self-reversing properties of rock-forming ferrimagnetic minerals could be demonstrated in the laboratory (Uyeda, 1958; Carmichael, 1959; Havard and Lewis, 1965), but some of the self-reversal mechanisms may not be reproducible on a laboratory time scale (Néel, 1951; Verhoogen, 1956).

Ade-Hall (1964) reported a fairly strong correlation between the polarity of natural remanent magnetization (NRM) and the stage of alteration or oxidation of minerals in basalt lavas of Mull, Scotland. Although paleomagnetic studies did not reveal systematic differences between normal and reversed lava flows (Wilson, 1964), Ade-Hall claimed on the basis of his petrological observations that a self-reversal mechanism might be responsible for the NRM of the reversed flows. Wilson and Watkins (1967) claim a correlation between petrography and polarity but agree that field reversals have occurred. Larson and Strangway (1966), alternatively, concluded that in the extrusive rocks studied by them there is no

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correlation between petrography and polarity.

It seems, therefore, important to further test the hypothesis of field reversal. Presently, the results which favor field reversals are: (1) Occurrence of the same magnetic polarity (normal or reversed) of NRM in different parts of the world for a certain time interval. (2) Similarity of petrographic properties between normal and reversed rocks. (3) Inability to produce partial or complete self-reversals in the laboratory. (4) Consistency in the directions of NRM between igneous and ajacent baked rocks. (5) Stability of NRM in rocks when tested by ac demagnetization, thermal demagnetization, storage test, etc.

Reliable information will be obtained about the behaviour of the past geomagnetic field if any geologically clear succession of lava flows is thoroughly investigated according to these criteria. The present paleomagnetic study of some volcanic sequences of Pliocene in the Southwestern United States has been carried out with this in mind. Paleomagnetic results for rocks of this age can be compared with those of the Hakone-Izu volcanic region in Japan (Nagata et al., 1957, 1963; Kono et al., 1964). Moreover, by means of radiometric ages magnetic polarities can be correlated with those observed in other parts of the world (Cox et al., 1965; McDougall et all., 1966).

2. Scope of Investigation

A sampling trip was made in June, 1965, to New Mexico and Arizona. From 24 basalt lava flows in the Rio Grands Gorge, near Taos, New Mexico, and near Flagstaff, Arizona, 164 oriented cores were obtained by means of a U.S.G.S. type core drill (Doell and Cox, 1965). Fourteen oriented hand samples were taken from four baked sedimentary layers in these sampling areas. The approximate direction of NRM was measured at the sampling site with a portable fluxgate magnetometer. It was possible in this way to determine the directions of NRM with an error of less than fifteen degrees. Hence, the polarity of rocks was easily distinguished and samples which had been affected by lightning strikes were avoided. Core samples were drilled with a horizontal interval of two to five meters, and if the flow was sufficiently thick, two or three vertical levels were sampled. Specimens, 2.4 cm in diameter and 2.3 cm long, were cut from these core samples and used in the paleomagnetic and radiometric dating experiments.

NRM of these specimens was measured by spinner magnetometers in U.S. Geological Survey, Menlo Park, and in Department of Geophysics, Stanford University within two weeks after sampling. At least two of the following stability tests were applied to each specimen: ac demagnetization, storage test, low-temperature demagnetization (Ozima et al., 1964), and thermal demagnetization by Thelliers' method (Thellier and Thellier, 1959). Most of these experiments were carried out in Kakioka Geophysical Laboratory, University of Tokyo, where the natural and artificial remanent magnetization was measured by an astatic magnetometer.

Potassium-Argon ages were determined at the Geophysical Institute, University of Tokyo, from specimens which were not heated in the paleomagnetic experiments. At least one polished and one thin section were made from each lava flow and petrographic observa-

tions were carried out on these sections. Thermomagnetic and X-ray analyses were also performed on magnetic minerals separated from each lava flow.

3. Geology, Petrology and Chronology of Lava Flows

3-1 Rio Grande Gorge, New Mexico

The Taos Plain is covered by a series of late Tertiary-Quaternary olivine basalt lavas, ranging from 5 to 20 meters in thickness, with interbedded coarse sediments (sands and gravels) which range in thickness from 5 to 30 meters (McKinlay, 1957). The lavas erupted from fissures and covered this area in a flat-lying succession. About 1000 feet of strata are well exposed in the Rio Grande Gorge. Core samples were taken on the western cliff of the Rio Grande, near the bridge on New Mexico State Highway 111 (Bridge Site) and at the confluence of the Rio Grande and the Rio Taos (Rio Taos Site, in Figure 1). At the Bridge Site a thick section of flows is well exposed, essentially unweathered, and complicated structurally by only one small fault. Fourteen lavas and two baked sediment layers were cored. At the Rio Taos Site three flows stratigraphically lower than the section at the Bridge Site were sampled. All of these lavas and sediments are represented in Figure 2. The levels of the subsequence layers (817, 818, 820, 828 and 834), which was sampled upstream from the main sequence, are replaced in this figure about 70 meters upward from the actual positions so as to show the original stratigraphical relations relative to the main

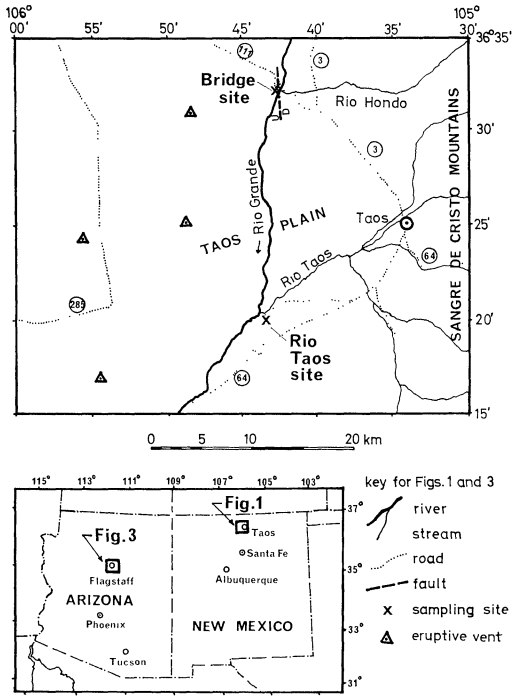


Fig. 1 Rio Grande sampling sites, with an index map for New Mexico and Arizona sampling localities.

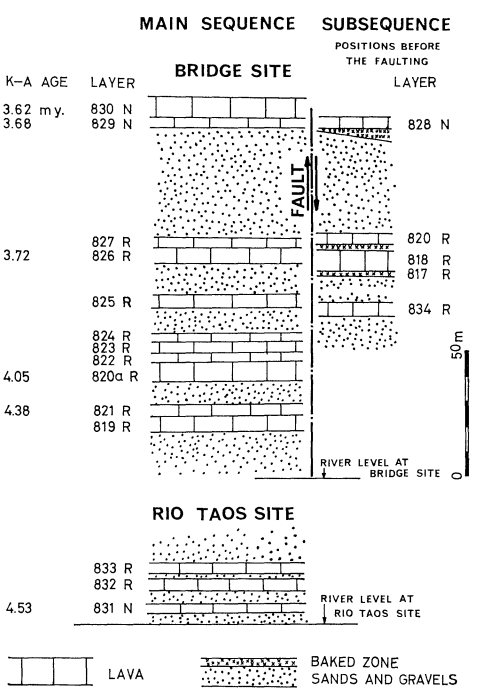


Fig. 2 Sampling section in the Rio Grande Gorge, New Mexico.

sequence before the faulting. Because of slight lateral stratigraphic variations the section in the subsequence does not exactly coincide with the main section.

All the lavas are olivine-rich basalts. Those from the Bridge Site are unweathered; they are generally composed of olivine (about 12–20 volume percent) sometimes with iddingsite rims, pyroxene (0–20%), opaques (1.5–5%), plagioclase (about 50%), and late crystallizing fractions, including small crystals of pyroxene, opaques and volcanic glass (5–22%). In spite of large amount of iron contained in these rocks, little is present in opaques. Table 1 lists the results of chemical analyses of these rocks by Dr. Aoki (personal communication, 1965). For a more complete discussion of the petrology of these rocks the reader is referred to Aoki (1967).

Table 1. The Chemical Composition of the Rio Grande Lavas

lava	SiO_2	TiO_2	Al_2O_3	Fe_2O_3	FeO	MnO	MgO	CaO	Na_2O	K_2O	H_2O^+	H_2O^-	P_2O_5	Total
830	51.04	1.28	16.56	1.91	8.62	0.17	6.75	8.75	2.80	0.60	0.87	0.11	0.16	99.62
829	50.02	1.08	16.28	3.05	8.62	0.16	7.62	8.61	2.73	0.64	0.82	0.17	0.14	99.94
827	48.00	1.56	16.50	3.59	8.88	0.18	7.79	9.53	2.81	0.59	0.36	0.14	0.21	100.14
826	51.04	1.62	17.02	0.93	8.59	0.14	5.88	7.78	3.36	1.66	0.86	0.30	0.33	99.51
824	51.51	1.49	17.47	3.68	5.72	0.09	5.76	7.74	3.41	1.67	0.66	0.22	0.32	99.74
820a	48.57	1.50	16.88	2.90	8.43	0.16	7.63	9.02	2.81	0.56	0.85	0.29	0.20	99.80
833	48.86	1.29	16.19	5.06	7.08	0.15	7.66	9.10	2.84	0.43	0.64	0.16	0.21	99.67
832	51.56	1.12	15.42	3.72	7.10	0.15	7.27	8.00	2.84	0.87	1.10	0.47	0.18	99.73
831	46.48	1.07	15.39	5.39	5.21	0.15	7.01	7.09	2.60	1.60	4.29	3.37	0.15	99.80
820	48.57	1.50	16.88	2.90	8.43	0.16	7.63	9.02	2.81	0.56	0.85	0.29	0.20	99.80
818	51.55	1.61	17.53	3.93	5.07	0.14	5.76	7.64	3.64	1.67	0.81	0.52	0.31	100.17

Samples from Rio Taos Site are generally very similar petrographically to those at the Bridge Site, except that most were weathered to some degree. In most cases, the olivine and volcanic glass were weathered to yellowish-green clay (montmorillonite).

In Figure 2, the ages of the lava flows are also given. Details of the chronological work has been reported in a separate paper (Ozima et al., 1967). The internal consistency between the ages and the stratigrophic order is very great. These flows represent a time interval of about one million years of the late Pliocene; yet no indication of weathered or eroded flow tops was observed in the field. Apparently the area was one of aggradation and streams flowing into the area were quickly covering the flow with sands and gravels.

3-2 Basalts near Flagstaff, Arizona

A total of seven basalt lavas and two baked sediment layers were sampled at three sites near Flagstaff, Arizona (Figure 3). These flows were classified as "older basalts" by Cooley (1962); They preceded the volcanic activities in the San Francisco mountains, north of Flagstaff, and are geologically dated as middle Pliocene. In Oak Creek Canyon, lava flows of 5–20 meters in thickness unconformably overlie Permian Coconino sandstone. Eroded or weathered interfaces were not found at our sampling sites, but Cooley (1962) reported the occurrence of such interfaces in other areas. Samples were also collected from

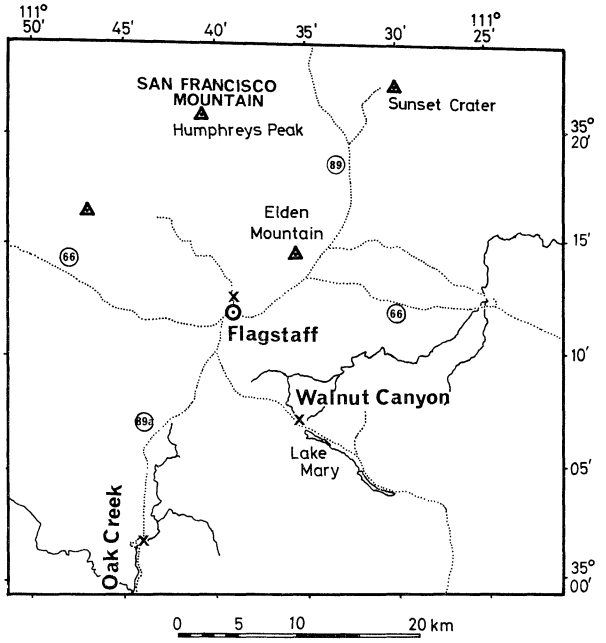


Fig. 3 Arizona sampling sites. The index map and the explanation of symbols for this figure are given in Fig. 1.

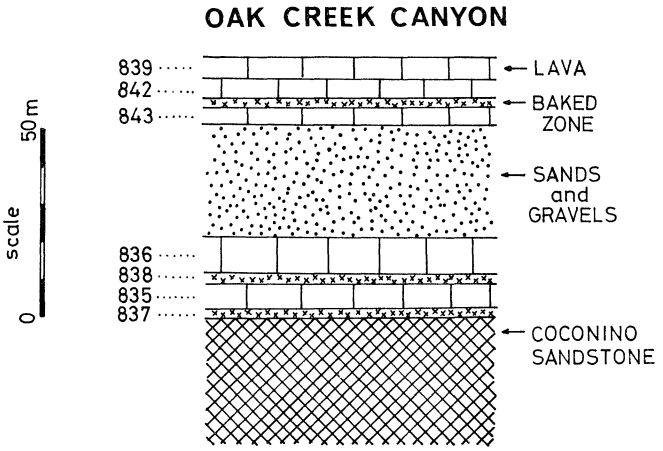


Fig. 4 Sampling section in Oak Creek Canyon, Arizona.

a basalt lava in Walnut Canyon and one on N.E. San Francisco Street, Flagstaff, for a test of magnetic stability. The stratigraphic section in Oak Creek Canyon is schematically shown in Figure 4.

These lavas are olivine (high alumina) basalts. The rocks are composed of olivine, plagioclase, Ca-pyroxene, opaques and fine-grained, late-crystallizing materials such as tiny opaque crystals and pyroxene. Although K-Ar dating was attempted, the presence of varying amounts of clay in these rocks made it impracticable.

4. Thermomagnetic Analyses, X-ray Analyses and Microscopic Observations of Ferrimagnetic Minerals

Thermomagnetic analyses were carried out on ferrimagnetic minerals separated from each basalt and baked sediment. The results for basaltic samples are very similar to each other, regardless of where collected or their magnetic polarity. Typical examples of the Js-T curves are shown in Figure 5. The saturation magnetization (Js) decreases smoothly on heating and disappears around 550°C, showing that the main ferrimagnetic minerals are titanomagnetites. On the cooling process, the Curie temperature (Tc) of some samples drops down by about fifty degrees and the Js decreases by 5 to 10%. According to Ozima and Larson (1967), these features can be explained by the reduction of titanomaghemite

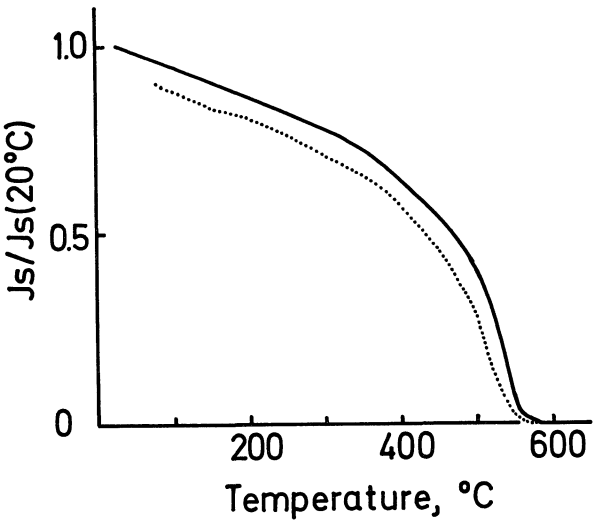


Fig. 5(a)

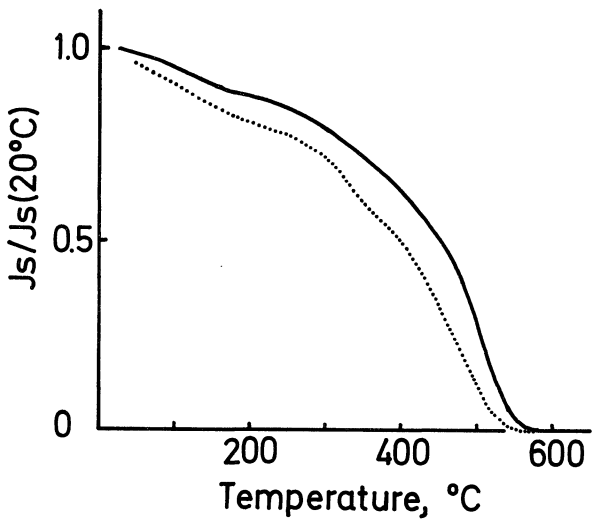


Fig. 5(b)

phases to titanomagnetite in the heating process. On the other hand, the thermomagnetic curves on baked sediments show a wide variety (Figure 6). The Js-T curve for sediment layer 837 (Figure 6 c) is similar to that for many basalts. However, in layer 817 (Figure 6 a), there is a range of Curie Temperature, the highest of which (670°C) may correspond to that of hematite. In layers 828 and 838 (Figure 6 b, d) the Tc is about 550°C, but the mode of

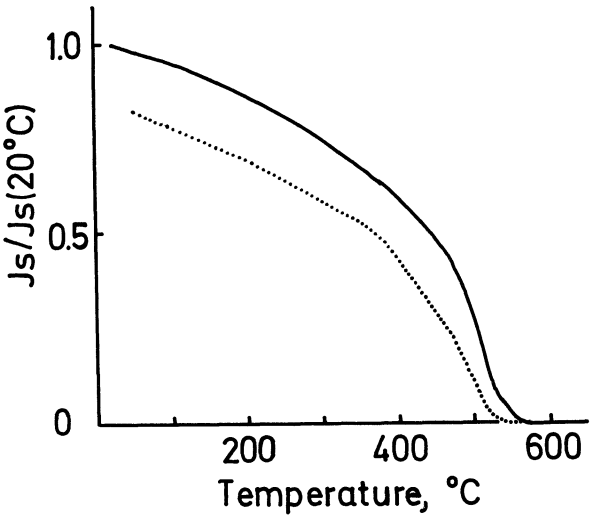


Fig. 5 (c)

Fig. 5 Changes of saturation magnetization versus temperature of ferrimagnetic minerals of basalts, measured in vacuum of 10^{-3} mmHg and with an external field of 5000 Oe.
(a) Sample: 823, Rio Grande, reversed.
(b) Sample: 829, Rio Grande, normal.
(c) Sample: 836, Oak Creek Canyon, reversed.
—— heating curve ····· cooling curve

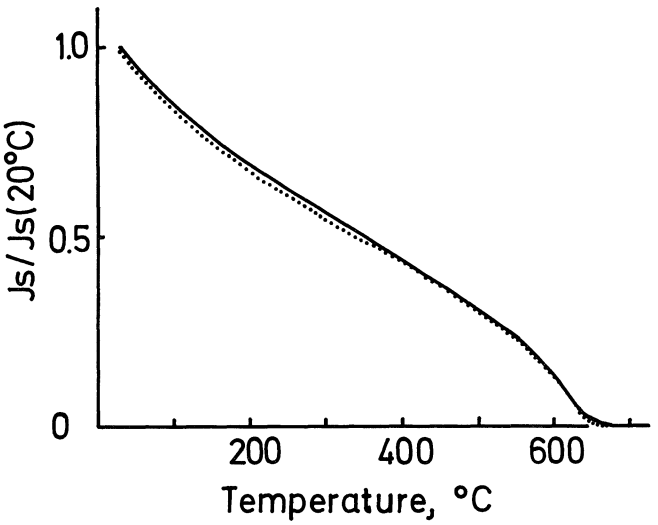


Fig. 6 (a)

decrease in Js on cooling is rather different from that of basalts, indicating different amounts of titanium and/or different ratios of titanomaghemite to titanomagnetite phases in these rocks.

X-ray diffraction analyses were done by means of NORELCO on these separated minerals. The diffraction curves of some samples show a definite split of the main peaks indicating the co-existence of at least two cubic phases with slightly different lattice constants. The one having the larger 2θ angle is probably oxidation product (titanomaghemite) of the other. Results of the thermomagnetic and X-ray analyses are listed in Tables 2 and 3. X-ray analyses of baked sediments, which contain little ferrimagnetic material, have been unsuccessful except for the layer 837.

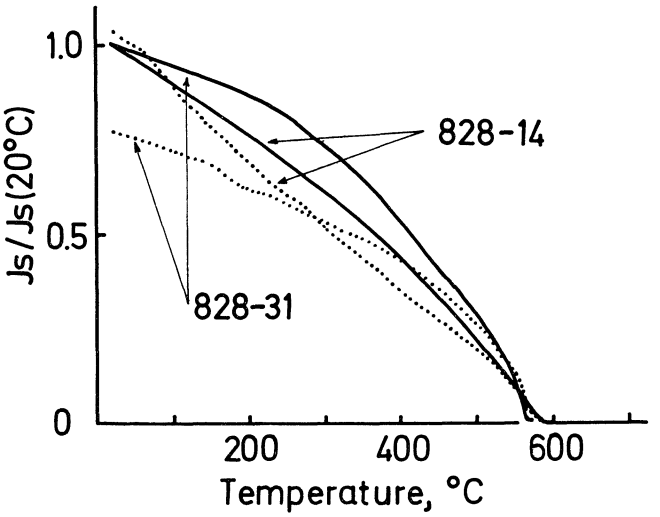


Fig. 6 (b)

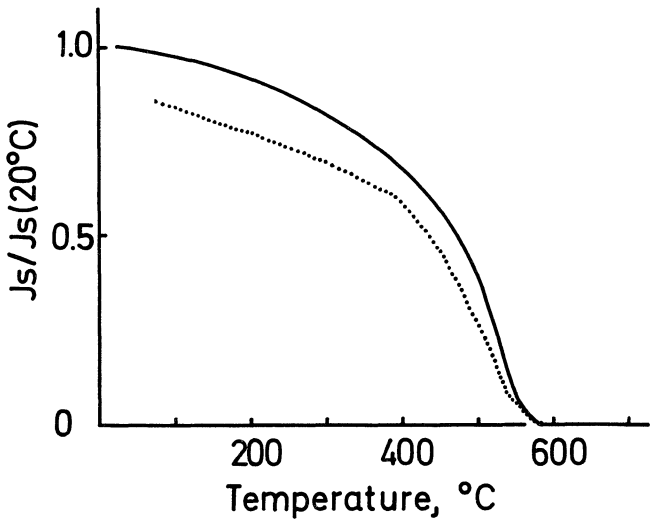


Fig. 6 (c)

All the ferrimagnetic minerals of the basalts were further examined in detail by observations of polished sections at magnification up to 2400X. Although the general petrography is very similar in all samples (section 3), the ferrimagnetic minerals (mostly titanium iron oxides) show various stages of oxidation. The results of these observations are also summarized in Tables 2 and 3. Figure 7 gives typical examples of these oxidation stages observed by reflected light under oil immersion. Wilson and Haggarty (1966) provide an excellent series of photos depicting this progressive oxidation. The occurrence of titanomaghemite (a low temperature oxidation of titanomagnetite) was ignored in this classification. As far as the present data are concerned, there is little correlation between the magnetic polatiry of NRM and the degree of the high-temperature oxidation of ferrimagnetic minerals. Neither self-reversal by petrological control (Ade-Hall, 1964) nor the correspondence of oxidation and polarity epochs (Wilson and Watkins, 1967) seems likely in the present specimens.

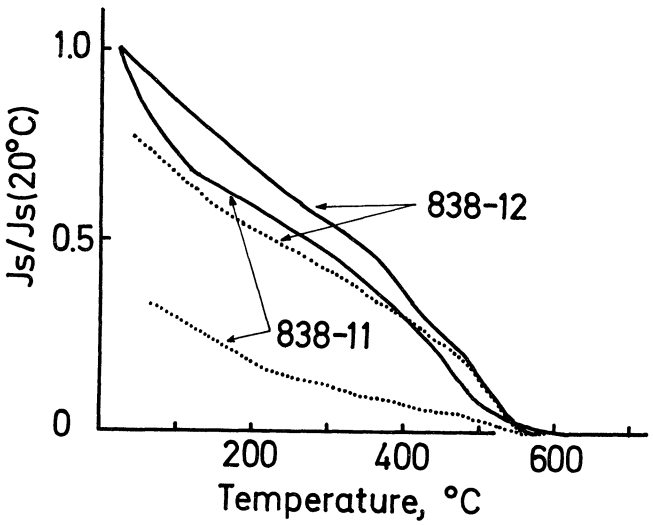


Fig. 6 (b)

Fig. 6 Changes of saturation magnetization versus temperature of ferrimagnetic minerals of baked sedimentary layers measured in vacuum of 10^{-3} mmHg and with an external magnetic field of 5000 Oe, if not otherwise stated.
(a) Sample: 817-13, Rio Grande, reversed, in air and 2850 Oe.
(b) Sample: 828, Rio Grande, normal, in air and 2850 Oe.
(c) Sample: 837-12, Oak Creek Canyon, reversed.
(d) Sample: 838, Oak Creek Canyon, reversed.
—— heatin curve ····· cooling curve

Table 2. Various Properties of Ferrimagnetic Minerals of Samples from Rio Grande

layer	NRM	a	Tc	sample	mtL	mtS	ilL	ilC	ht	oxid.
				Main Sequence						
830	N	8.380	545	830-11	2	3	2	1	3	H
829	N	8.435 8.383	545	829-13	3	3	4	2	3	H
827	R	8.397	560	827-11	3	3	4	2	3	H
826	R	8.390	550	826-11	3	3	3	2	3	H
825	R	8.409	545	825-15	2	3	3	2	3	H
824	R	8.432	535	824-13	2	3	3	2	3	H
823	R	8.381	550	823-13	1	3	0	0	1	N
822	R	8.427	560	——						
820a	R	8.397	560	820-23	2	3	0	2	1	N
821	R	8.407	555	821-11	2	3	4	2	2	MH
819	R	8.400 8.451	560	819-13	3	3	3	2	2	MH
833	R	8.457	520	833-11	3	3	0	2	1	N
832	R	8.411 8.459	550	832-12	3	3	3	2	3	H
831	N	8.388	550	831-11	3	3	0	1	1	N
				Subsequence						
828*	N	——	580	——						
820	R	8.397	560	820-11	3	3	3	2	4	VH
818	R	8.395	545	818-11 818-33	3 3	3 3	3 0	2 1	2 1	MH N
817*	R	——	670	——						
834	R	8.397 8.427	545	834-11	3	3	0	1	1	N

1. lava: number of lava or baked sediment layer, with an asterisk (*) for the latter case.
2. NRM: magnetic polarity of NRM, N for normal and R for reversed.
3. a: lattice constant(s) in Å.
4. Tc: Curie temperature in °C.
5. sample: number(s) of sample(s) from which the polished section(s) was made.
6. mtL: amount of large (diameter >10 microns) crystals of titanomagnetite.
7. mtS: amount of tiny crystals of titanomagnetite (late crystallizing fraction).
8. ilL: amount of ilmenite lamellae exsolved in (lll) planes of titanomagnetite.
9. ilC: amount of single crystals or band-like crystals of ilmenite.
10. ht: amount of hematite.
11. oxid.: degree of the high-temperature oxidation of ferrimagnetic minerals.

The abundance of materials (columns 6-10) is listed by numbers 0, 1, 2, 3 or 4 which stand for;

0: not present,
1: rare or few,
2: common,
3: abundant,
4: very abundant,

The degree of high-temperature oxidation (column 11) was defined by one of us (E.E.L.) in the following manners (cf. Buddington and Lindsley, 1964);

No oxidation (N): uniform spinel-cubic phase only.

Moderately high oxidation (MH): ilmenite lamellae present, a little hematite present on the rims and along cracks of titanomagnetite.

High oxidation (H): ilmenite lamellae abundant, hematite common to abundant, metailmenite present.

Very high oxidation (VH): very abundant hematite, commonly all titanomagnetite oxidized to rutile (TiO_2), hematite, and/or pseudobrookite ($TiFe_2O_5$).

Table 3. Various Properties of Ferrimagnetic Minerals of Samples from Oak Creek Canyon, Walnut Canyon and Flagstaff**

layer	NRM	a	Tc	sample	mtL	mtS	ilL	ilC	ht	oxid.
Oak Creek Canyon										
839	R	8.394 8.422	535	839-11	3	3	2	2	3	H
842	R	8.380	530	842-11	3	3	2	2	4	V H
843	R	8.397	540	843-11	3	3	2	2	4	V H
836	R	8.388	530	——						
838*	R	——	555	——						
835	R	8.395	540	——						
837*	R	8.382	565	——						
Walnut Canyon										
840	R	8.396	555	840-11	3	3	4	3	4	V H
Flagstaff City										
841	R	8.370	560	841-11	3	3	4	3	3	V H

** explanation of symbols and abbreviations is given in Table 2

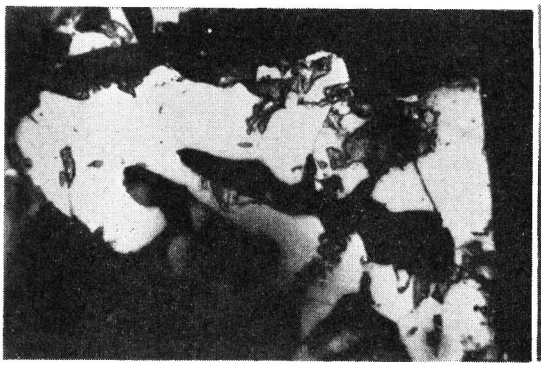


Fig. 7(a) 818-33

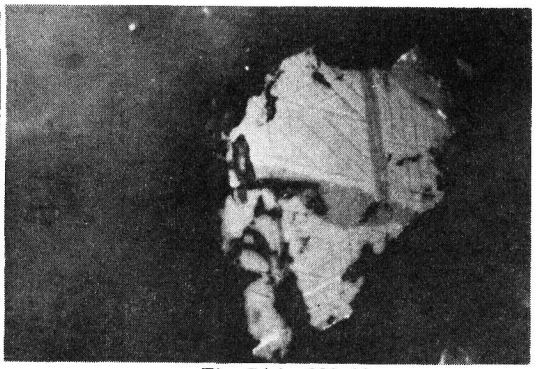


Fig. 7(c) 830-11

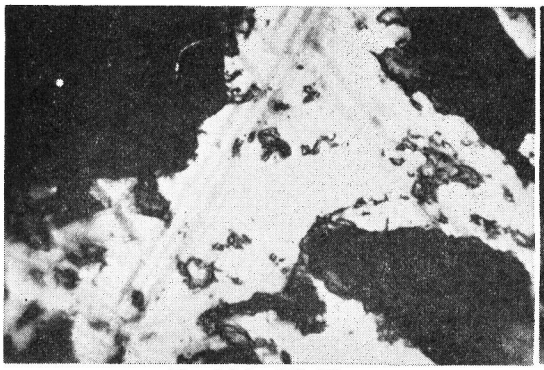


Fig. 7(b) 819-13

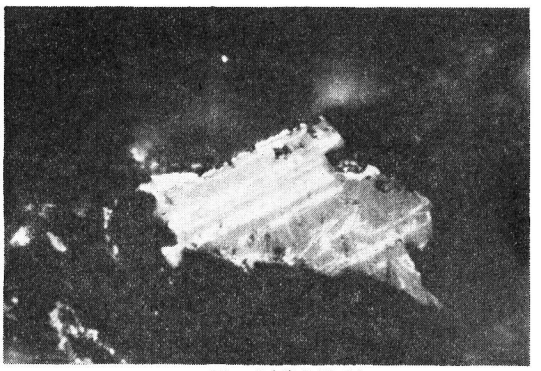


Fig. 7(d) 843-11

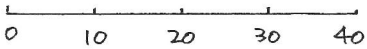


Fig. 7 Photomicrographs of changing degrees of high-temperature oxidation in titanomagnetite.
(a) Sample 818-33 showing uniform grains of Ti-rich magnetite, no oxidation (N).
(b) Sample 819-13 showing the ilmenite exsolution in (111) planes of magnetite, moderately high oxidation (MH).
(c) Sample 830-11 showing abundant ilmenite lamellae and hematite, high oxidation (H).
(d) Sample 843-11 showing hematite, metailmenite and pseudobrookite produced by very high oxidation (VH).

5. Direction and Stability of NRM

5-1 Direction of NRM in baked contacts

NRM of the sedimentary layers is usually very weak ($\sim 10^{-6}$ emu/g). However, within a few tens of centimeters below the four lava flows, the magnetization is almost as strong as that of basalts (10^{-4}–10^{-3} emu/g) and has directions that are consistent with the overlying flows (Figure 8). It is probable that these zones acquired thermo-remanent magnetization (TRM) when the lavas flowed over the sediments and heated the surface layers. Commonly, the heated surfacial zone is reddened but in places no such reddening occurs. At the Bridge Site, not only the specimens from the reddish part but also specimens collected from white parts of the contact sediment show directions of NRM concordant with those in the lavas. The consistency of NRM directions of the various sediments with those of the overlying basalt, despite wide variation in Tc and slopes of the Js-T curves, is a strong support for field reversals, and it also shows that NRM is stable in these rocks.

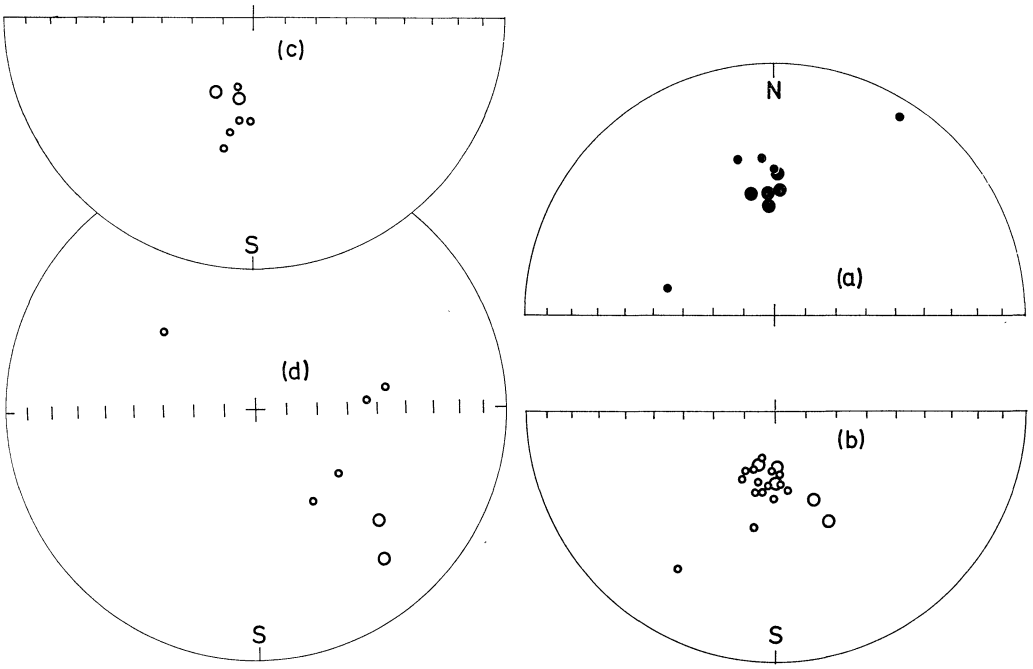


Fig. 8 Directions of NRM in baked contacts. Larger circles are from the baked sedimentary layers and smaller circles from the overlying basalt lavas. Solid symbols on the lower hemisphere and open circles on the upper hemisphere, equal area projection.
(a) 829 (lava) and 828 (baked sediment).
(b) 818 (lava) and 817 (baked sediment).
(c) 836 (lava) and 838 (baked sediment).
(d) 835 (lava) and 837 (baked sediment).

5-2 Storage tests and low-temperature demagnetization

After the measurements of NRM, the core specimens were stored with random orientations in the geomagnetic field. After 90 to 200 days, the magnetization of each specimen was again measured. Although the intensity remained almost unchanged (Table 4), some

Table 4. Intensity Change in the Storage Test

sample of statistic	No. of samples	mean of *I/Io*	standard deviation	variance
Rio Grande reversed	108	1.011	0.140	0.01970
Rio Grande normal	17	0.959	0.094	0.00889
Arisona reversed	50	1.052	0.216	0.04642
All data combined	175	1.018	0.163	0.02656

Table 5. Intensity Change in the Low Temperature Demagnetization

sample of statistic	No. of samples	mean of *I/Io*	standard deviation	variance
Rio Grande reversed	97	0.936	0.150	0.02233
Rio Grande normal	13	0.915	0.134	0.01808
Arizona reversed	44	0.934	0.083	0.00695
All data combined	154	0.934	0.132	0.01730

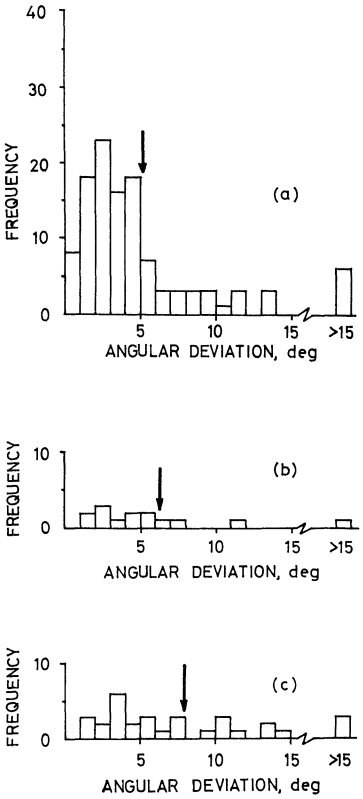


Fig. 9 Histograms of angular deviations of magnetization observed after storage of 90–200 days. Thick arrows indicate the mean values.
(a) Rio Grande reversed specimens.
(b) Rio Grande normal specimens.
(c) Arizona reversed specimens.

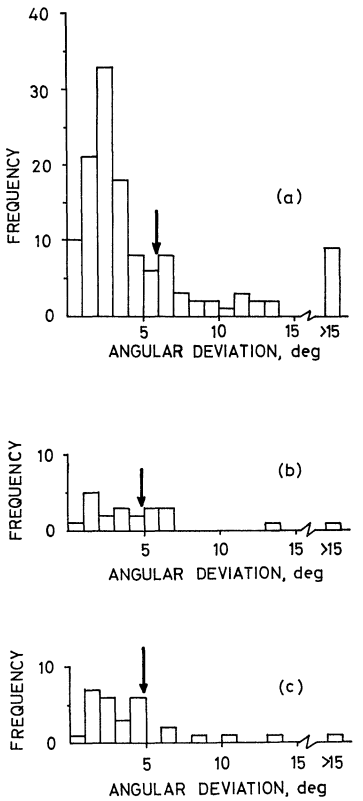


Fig. 10 Histograms of direction changes of magnetization observed in low temperature demagnetization.
(a) Rio Grande reversed specimens.
(b) Rio Grande normal specimens.
(c) Arizona reversed specimens.

specimens showed considerable changes in direction of NRM (Figure 9). The specimens were cooled to liquid nitrogen temperature, −196°C, and heated to room temperature in zero magnetic field. Changes in intensity and direction of magnetization are summarized in Table 5 and Figure 10, respectively. The principle of this method (Ozima et al., 1964) is that the soft part of remanent magnetization can be demagnetized at the temperature where the crystalline anisotropy energy becomes zero (−143°C for magnetite). With increase of Ti content in the titanomagnetite, the zero crystalline-energy temperature depresses. In basic rocks the amount of Ti is commonly great enough to shift the temperature below that of liquid nitrogen (Syono and Ishikawa, 1963). As many of our samples contain rather large amounts of Ti, the change by this treatment is not very significant.

The resemblance of Figures 9 and 10 is only accidental, as the direction changes in the two treatments (storage and low-temperature) for the same specimen seem to be independent to each other (Figure 11). The responses of the normal and reversed samples from the Rio Grande to either test are grossly similar and this is also consistent with the view that there is no difference in magnetic properties between these two groups.

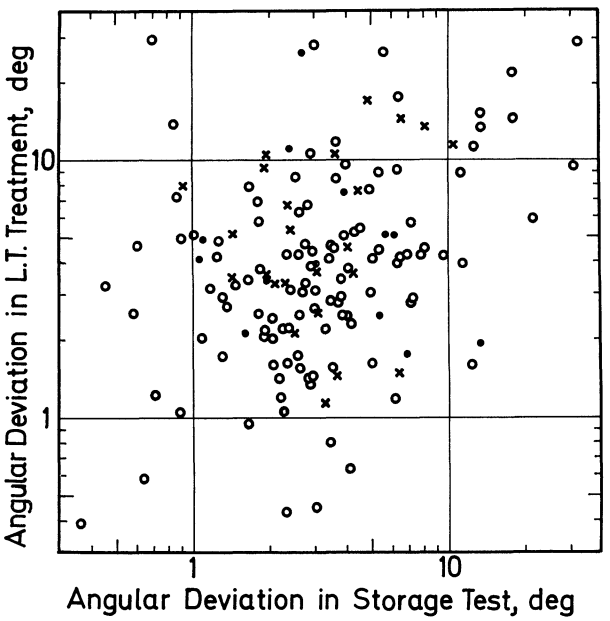
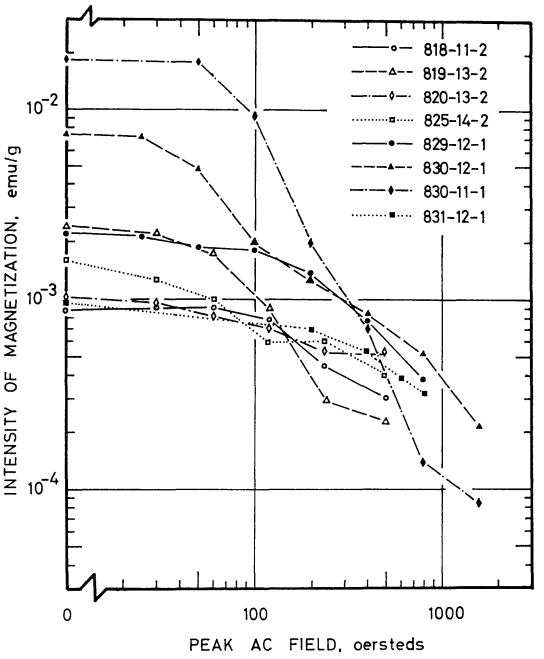


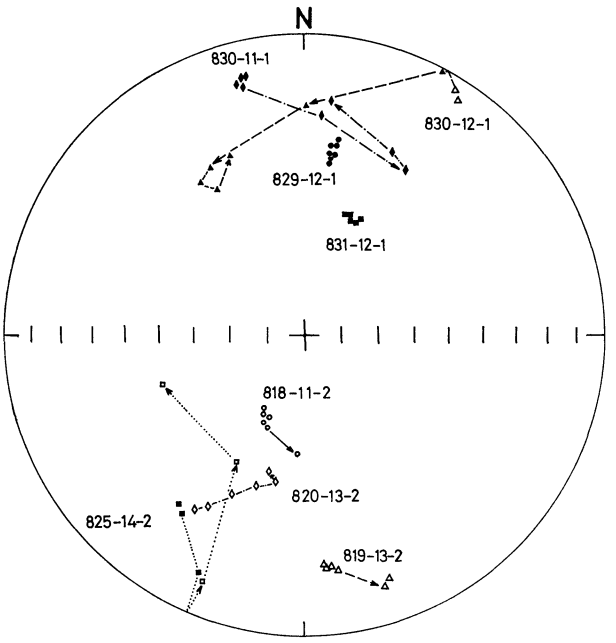
Fig. 11 Relation between the direction changes observed in storage test and low temperature demagnetization. Open circles for Rio Grande reversed specimens, dots for Rio Grande normal and crosses for Arizona reversed.

5-3 AC demagnetization

About two specimens from each lava flow were demagnetized in a stepwise manner in gradually decreasing alternating magnetic fields. Examples of typical changes for Rio Grande specimens are shown in Figure 12. Some specimens have large components of soft magnetization and lose more than fifty per cent of their original NRM in a field of 100



(a) Intensity change.



(b) Direction change, shown in the same manner as in Fig. 8.

Fig. 12 Examples of ac demagnetization for Rio Grande specimens.

oersteds. In some of these specimens the direction of magnetization also greatly changes. Such magnetically soft behaviors have been observed in about thirty per cent of the ac demagnetized specimens, both for normal and reversed. But in all cases, the k values of

Fisher's (1953) statistics increased after the ac demagnetization. Thus the difference between "soft" and "hard" samples is mainly in the amount of the soft component. The soft component can be effectively removed by ac demagnetization. As shown in Figure 12, there is no significant difference in stability between the normal and reversed groups.

5-4 Thermal demagnetization

Stepwise demagnetization (Thellier and Thellier, 1959) was applied to at least one sample from each lava flow (Kono and Nagata 1966). The detailed report of this experiment will be published elsewhere (Kono and Nagata, 1968). Many of the samples yielded reliable measures of the intensity of the paleogeomagnetic field and that the directions of NRM remained almost unchanged after partial thermal demagnetization. These results also indicate that the NRM of these samples is stable and, moreover, there is no difference in behaviors between normal and reversed samples.

5-5 Summary of paleomagnetic experiments and virtual pole positions

Tables 6 and 7 summarize the results of the paleomagnetic experiments. Stable directions were determined after the application of the stability tests. Samples which showed angular changes of more than ten degrees in the storage test were discarded. These tables also list the dispersion factors defined by Fisher (1953) for the directions of the original

Table 6. Summary of Paleomagnetic Results, Rio Grande

layer No.	original NRM						stable component					virtual pole			
	N	I	D	k	α_{95}	J	N	I	D	k	α_{95}	lat.	long.	dp	dm
Main sequence															
830	6	26	4	6	31	1650	5	39	19	14	21	68	21	15	25
829	6	39	350	7	28	241	6	44	350	10	23	76	116	18	28
827	5	−44	177	176	6	131	5	−44	178	219	5	−79	264	4	6
826	10	−36	179	8	18	167	9	−43	188	46	8	−77	220	6	9
825	10	−59	228	4	26	227	9	−63	214	13	15	−63	136	19	24
824	10	−51	161	4	30	401	8	−48	163	17	14	−74	321	12	18
823	10	−53	157	15	13	271	8	−50	168	60	7	−79	318	6	10
822	10	−45	168	16	12	131	9	−46	161	43	8	−71	321	6	10
820a	10	−48	166	164	4	153	9	−50	165	329	3	−76	323	2	4
821	9	−48	169	183	4	78	9	−48	169	322	3	−79	308	2	4
819	8	15	120	2	50	65	8	−6	136	2	57	reversed?		—	—
833	2	−31	160	—	—	35	2	−34	162	—	—	−66	300	—	—
832	10	−67	165	79	5	121	9	−67	166	98	5	−73	43	7	9
831	5	62	8	11	23	92	5	55	28	208	5	67	339	5	7
Subsequence															
828*	5	51	357	290	4	85	—	—	—	—	—	—	—	—	—
820	5	−47	213	11	24	101	5	−50	203	38	12	−70	174	11	17
818	16	−65	200	11	11	186	13	−65	290	114	4	−77	104	5	6
817*	5	−64	168	44	11	33	—	—	—	—	—	—	—	—	—
834	7	−8	155	3	41	92	5	−19	164	19	18	−60	287	10	19

Table 7. Summary of Paleomagnetic Results, Arizona

layer No.	original NRM						stable component					virtual pole			
	N	I	D	k	α_{95}	J	N	I	D	k	α_{95}	lat.	long.	dp	dm
Oak Creek Canyon															
839	5	−61	223	15	20	254	5	−64	216	19	18	−61	125	23	28
842	5	−59	173	134	7	219	5	−62	182	210	5	−82	76	6	8
843	5	−31	176	245	5	311	5	−31	176	391	4	−71	260	2	4
836	5	−55	189	95	8	88	5	−54	188	114	7	−83	161	7	10
838*	2	−64	198	—	—	42	—	—	—	—	—	—	—	—	—
835	5	−68	100	5	37	320	4	−53	128	21	20	−47	352	19	28
837*	2	−65	170	—	—	7	—	—	—	—	—	—	—	—	—
Walnut Canyon															
840	5	−35	188	183	6	170	5	−37	187	228	5	−74	224	3	6
Flagstaff City															
841	5	−57	335	2	82	484	5	−60	327	2	79	reversed?		—	—

* baked sedimentary layers
N: numbers of original/stable samples.
I : inclination in degrees, downward positive.
D: eastward declination from north, in degrees.
k : Fisher's precision parameter.
α_{95}: semiangle of cone of 95% confidence for mean direction, in degrees.
J : average intensity of original NRM in 10^{-5} emu/g.
lat.: latitude of virtual geomagnetic pole, positive for northern latitude.
long.: eastward longitude of virtual geomagnetic pole, in degree.
dp, dm: semiaxis of oval of 95% confidence for virtual geomagnetic pole.

NRM and the stable component. Virtual geomagnetic pole positions are determined with the dipole assumption, from the directions of the stable component (Figure 13).

6. Conclusions

The results of the present investigation are clearly in favor of the hypothesis of field reversal.

(1) Samples from the contact sedimentary zones have the same directions of NRM as those of the overlying lavas.

(2) Stability of NRM of both normal and reversed samples is relatively high and shows similar scatter.

(3) No correlation is found between the polarity of NRM and the magnetic and petrographic properties of the ferrimagnetic minerals contained in the rocks.

(4) The ages of the reversals found in the present study seems to be consistent with the data from other parts of the world (Cox, Doell and Dalrymple, 1965;McDougall et al., 1966).

It may thus be concluded that the directions of the NRM of rocks obtained in the present investigation represent those of the geomagnetic field in the Southwestern U.S.A. when the lavas were erupted. The virtual geomagnetic poles from Rio Grande and Arizona

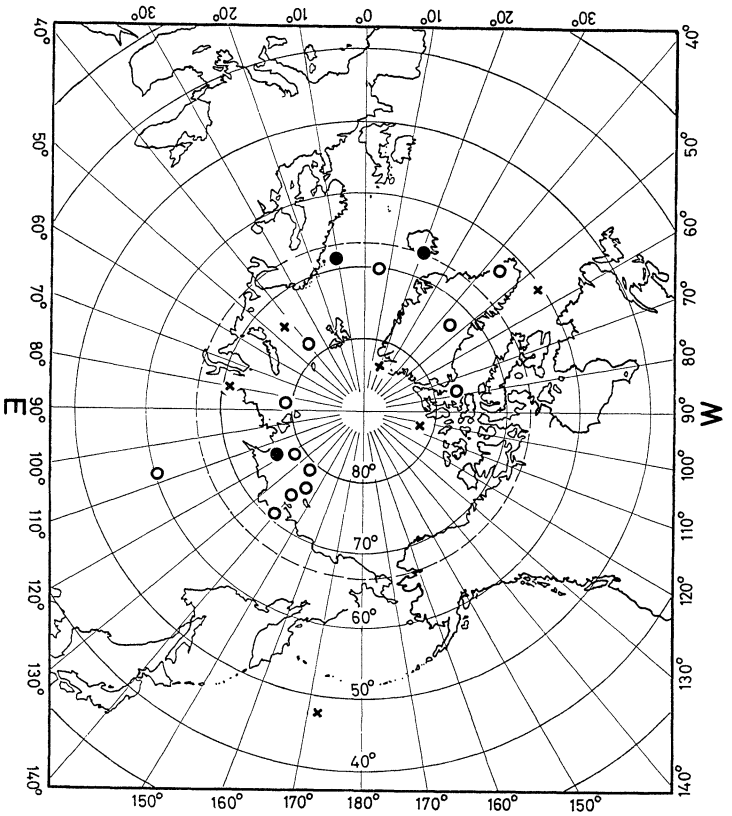


Fig. 13 Virtual geomagnetic pole positions for late Pliocene in the Southwestern U.S.A. Open circles from Rio Grande reversed lavas, dots from Rio Grande normal and crosses from Arizona reversed.

samples, which are near to the present geographic pole (Figure 13), are also in good agreement with other paleomagnaetic data from the Pliocene.

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